

MCQ:**6*1 =6 marks**

1. In a circular doubly linked list with a dummy head node, what node does head.prev.next refer to?
A) The second last node in the list B) The dummy head node C) The last node
2. Suppose a 3-D array has dimensions $4 \times 4 \times 5$. Find the linear index corresponding to the 3-D index [3][3][1] using row-major order. Assume 0-based indexing.
A) 76 B) 56 C) 66
3. Which type of linked list has no NULL pointers in any node?
A) Singly linked list B) Doubly linked list C) Circular linked list
4. When inserting a new element at the end of a doubly linked list, how many pointer references need to be modified?
A) 2 B) 3 C) 4
5. Suppose the dimension of a 3-D array is $4 \times 4 \times 5$. If the linear index is 37, find the corresponding 3-D index .(Assume 0-based indexing.)
A) [1][3][2] B) [1][2][2] C) [1][3][1]
6. In a circular doubly linked list, how can you directly access the last node without traversing the entire list?
A) head.next B) head.prev C) head.elem

7. Write a function/method named **compareRowSum(head, matrix)** that takes the head of a singly linked list and a two-dimensional matrix (2D array) as parameters. The function should calculate the sum of each row of the matrix in row-wise order and compare these sums sequentially with the values stored in the linked list. If all the linked list elements exactly match the corresponding row-wise sums of the matrix, the function should return **Matched**; otherwise, if any mismatch occurs, the function should return **Not Matched**. Assume that the number of nodes in the linked list is equal to the number of rows in the matrix.

Note: you are not allowed to create a new linked list/array.**[12marks]**

Example 1	Example 2																		
matrix= <table><tr><td>1</td><td>3</td><td>5</td></tr><tr><td>2</td><td>6</td><td>8</td></tr><tr><td>7</td><td>4</td><td>9</td></tr></table> Singly Linked list: 9→16→20 In this scenario function/ method will return Matched	1	3	5	2	6	8	7	4	9	matrix= <table><tr><td>1</td><td>3</td><td>5</td></tr><tr><td>2</td><td>6</td><td>8</td></tr><tr><td>7</td><td>4</td><td>9</td></tr></table> Singly Linked list: 9→15→20 In this scenario function/ method will return Not Matched	1	3	5	2	6	8	7	4	9
1	3	5																	
2	6	8																	
7	4	9																	
1	3	5																	
2	6	8																	
7	4	9																	

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public static string compareRowSum(Node head, int [][] matrix){    ||    def compareRowSum(head, matrix):
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